LEDGERCORE
ENTERPRISE MINING & CONCESSION EDITION

INSTITUTIONAL & CONCESSION GRADE

TECHNICAL PROPOSAL

Integrated Financial, Asset & Operations Engine for Mining Enterprises

An institutional-grade ERP platform architected specifically for high-capital extraction, fleet-intensive operations, and strict CEMAC / SYSCOHADA compliance.

- CEMAC / OHADA AUDCIF Revised
- Heavy Equipment & Fixed Plants
- Telematics & Operating Hour Maintenance
- Multi-Axial Analytical Budgeting
- Offline Concession Edge-Sync
- Dual IFRS 6 / IFRS 16 Reporting

CLIENT SECTOR	REGULATORY FRAMEWORK	ARCHITECTURE MODEL	CLASSIFICATION
Mining & Extraction	CEMAC / OHADA / IFRS	Distributed Cloud & Edge	Commercial-in-Confidence

01 Executive Summary & Industry Strategic Fit

Mining corporations operating within Central and West Africa (CEMAC / OHADA zone) encounter a unique convergence of operational and financial pressures. Mining assets are highly capital-intensive, extraction pits are geographically isolated with fragile communications infrastructure, equipment downtime costs tens of thousands of dollars per hour, and host-nation regulations mandate strict compliance with revised **SYSCOHADA standards**, local content laws, and specific mining royalty withholdings.

🚨 The Industry Dilemma: Fragmented Systems

Traditional mining sites rely on disconnected point solutions: standalone workshop spreadsheets, siloed procurement databases, and separate corporate accounting software.

- ✓ **Uncontrolled Budget Overruns:** Parts ordered for heavy machinery without real-time credit availability verification.
- ✓ **Distorted Asset Valuation:** Complex equipment overhauls expensed improperly rather than capitalized under componentized accounting.
- ✓ **Audit & Fiscal Penalties:** Late or non-compliant SYSCOHADA financial reporting and withholding tax liabilities.

✨ The LedgerCore Solution: Unified Convergence

LedgerCore delivers an institutional enterprise ecosystem where accounting, heavy fleet lifecycle, predictive maintenance, and multi-axial budgeting operate as a **single, synchronized organism**.

- ✓ **Real-Time Ceiling Enforcement:** Maintenance requisitions automatically check and commit budget envelopes before purchase orders are issued.
- ✓ **Units of Production (UOP) Depreciation:** Machinery depreciated based on engine hours or extracted tonnage.
- ✓ **Autonomous Edge Sync:** Concession pits operate 100% offline and reconcile seamlessly with HQ upon link restoration.

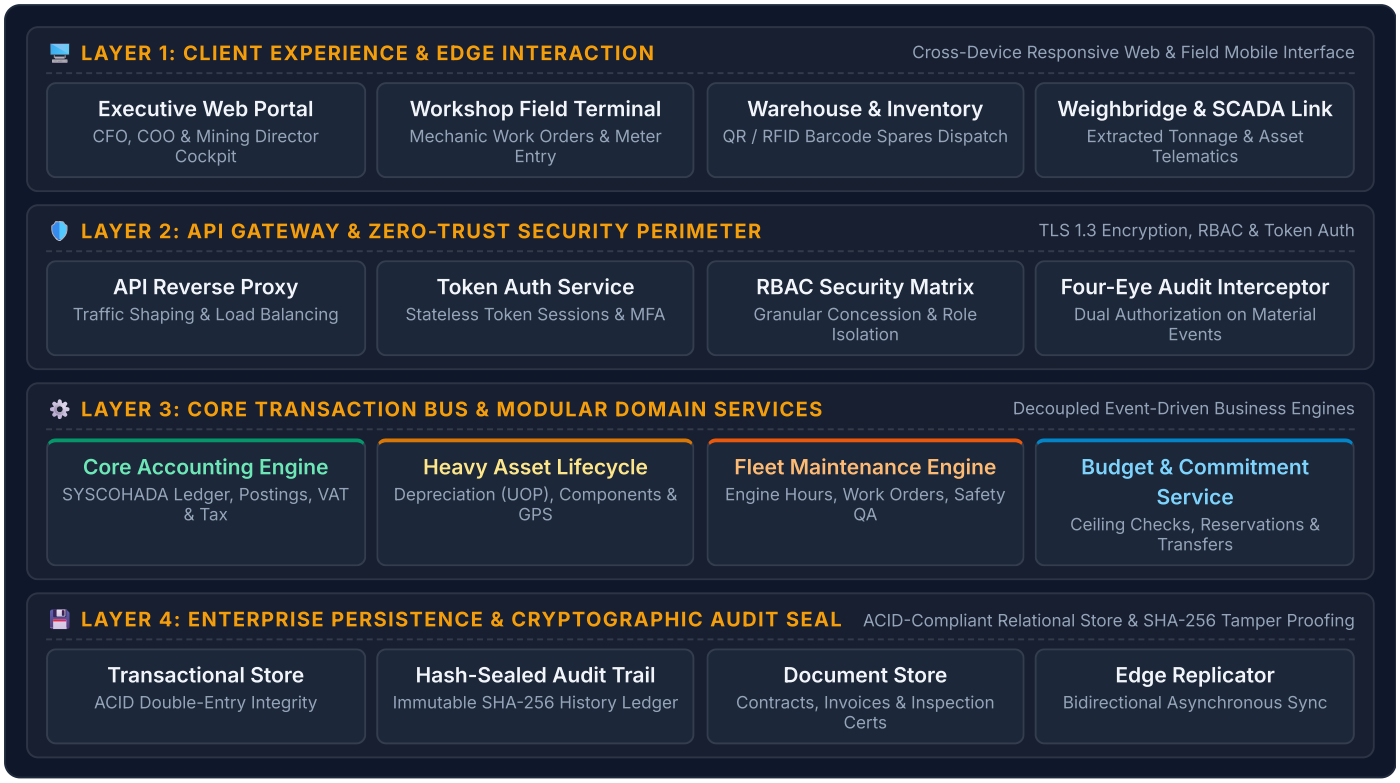
Core Operational Metrics Targeted

FLEET AVAILABILITY (OEE)	BUDGET VARIANCE	AUDIT RECONCILIATION	CEMAC COMPLIANCE
+28.5%	**0.00%**	**Real-Time**	**100% AUDCIF**
Preventive meter-based service triggers	Hard automated ceiling blocking	Automated double-entry posting engine	Mining royalties & fiscal seals



02 Enterprise Technical Architecture

LedgerCore is built on a **Distributed Micro-Service & Modular Tiered Architecture** designed for high availability, sub-second latency, zero transactional loss, and resilient operation across remote extraction zones. The architecture strictly decouples the user experience layer, business rule orchestration, security perimeter, and the financial transactional persistence store.



🛰️ Concession Edge-Sync: Zero-Interruption Offline Pit Operation

Remote extraction pits frequently suffer from satellite latency or fiber dropouts. LedgerCore employs an **Offline-First Edge Architecture**. Field supervisors can log work orders, dispatch spare parts, and verify pre-shift safety inspections locally without internet connectivity.

When satellite/GSM link is re-established, the Concession Edge-Sync automatically streams queued transactions using conflict-free deterministic resolution, ensuring headquarters has exact real-time financial visibility without halting pit operations.

🔒 Institutional Security & Four-Eye Segregation

Mining operations demand rigorous governance to prevent misappropriation of high-value parts, fuel, and capital expenditures. LedgerCore implements:

- ✓ **Granular Role-Based Access Control (RBAC):** Concession mechanics cannot modify purchase orders; accounting clerks cannot alter maintenance schedules.
- ✓ **Dual Authorization (Four-Eye Policy):** Credit transfers and asset write-offs require secondary supervisor approval.
- ✓ **Immutable SHA-256 Journal Sealing:** Fiscal period closures produce cryptographically sealed blocks that cannot be altered retroactively.

03 CEMAC / SYSCOHADA Core Accounting & Compliance

LedgerCore is built from the ground up to comply with the **SYSCOHADA Revised Accounting Framework (Acte Uniforme relatif au Droit Comptable et à l'Information Financière - AUDCIF)** governing all member states across the CEMAC and UEMOA economic communities. It combines statutory local compliance with international dual-standard reporting.

SYSCOHADA CHART OF ACCOUNTS

Native support for Classes 1 to 8 with specialized mining mappings: Class 2 (Exploration rights, concession development, stripping assets), Class 6 (Extraction reagents, royalties, rehabilitation provisions), Class 7 (Ore & concentrate sales).

DUAL REPORTING (OHADA + IFRS)

Simultaneous presentation of statutory OHADA financial statements alongside international IFRS adjustments: **IFRS 6** (Exploration & Evaluation), **IFRS 16** (Heavy Lease Capitalization), and **IAS 37** (Site Environmental Decommissioning).

MULTI-CURRENCY TREASURY

Functional currency in **XAF (Central African CFA Franc)** with automated FX revaluation on USD, EUR, and GBP. Essential for international ore export contracts, fuel import hedging, and overseas equipment procurement contracts.

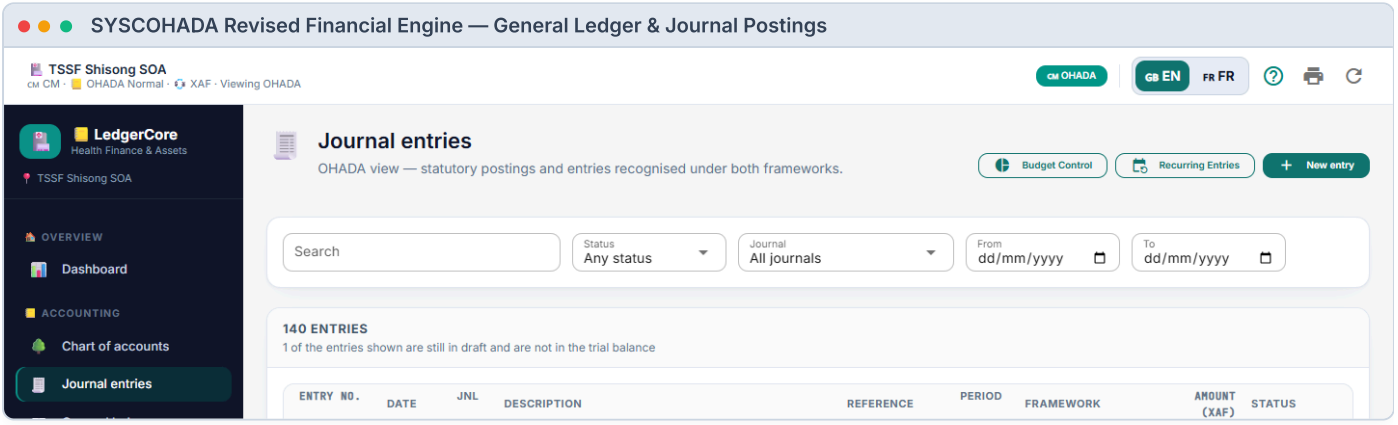
🏛️ CEMAC Tax, Mining Royalties & Statutory Withholdings Engine

Mining operators face complex withholding obligations under CEMAC fiscal directives. LedgerCore automates calculation, deduction, reporting, and voucher generation for:

TSR (Taxe Spéciale / Non-Resident)
Automated 15% withholding on technical assistance, drilling consultants, and foreign mining contractors.

Mining Royalties (Redevance Minière)
Automated deduction tied directly to extracted ore grade and export FOB market value.

Local Vendor Withholding
Automated 5.5% / 2.2% advance withholding tax with automated generation of official fiscal retention certificates.



04 Heavy Mining Asset Management

In mining operations, fixed assets represent the core of the balance sheet. LedgerCore manages the entire lifecycle of high-capital yellow machinery, fixed processing infrastructure, exploratory drill rigs, and site utility networks with industrial precision.

Fleet & Heavy Plant Governance

Complete custodial tracking for excavators (Caterpillar, Komatsu, Liebherr), 100-ton haul trucks, blast-hole drill rigs, primary jaw crushers, ball mills, and diesel generator farms.

- ✓ **Componentized Asset Accounting:** Compliant with IAS 16 & SYSCOHADA. The chassis, engine, and undercarriage of a haul truck are tracked as distinct components with separate depreciation schedules.
- ✓ **Rugged Asset Identification:** Support for industrial metal QR tags, RFID tags, and serial barcoding for rapid field auditing in harsh pit environments.
- ✓ **Location & Pit Assignment:** Real-time tracking of asset assignment across concessions, active pits, maintenance bays, and standby reserves.

Units of Production (UOP) Depreciation

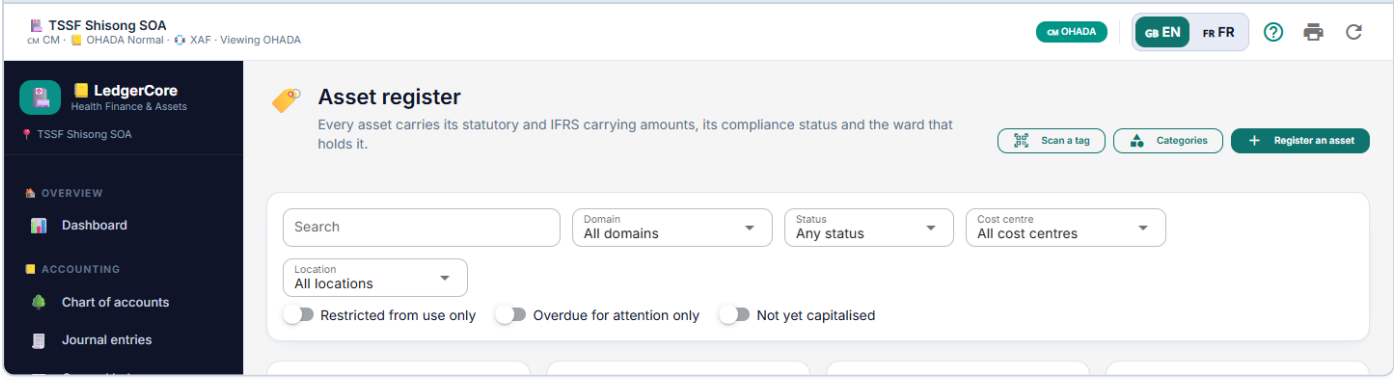
Traditional time-based straight-line depreciation misrepresents heavy mining assets, as wear-and-tear correlates directly with operating intensity rather than calendar months.

Mining-Specific UOP Formula:

```
Depreciation = (Asset Cost − Salvage Value) ×
(Hours Run in Period / Total Rated Engine Life)
```

LedgerCore supports **Units of Production (Engine Operating Hours or Mined Tonnage)**, Accelerated Reducing Balance, and Straight-Line methods, updating Net Book Value (NBV) automatically every billing cycle.

Heavy Asset Register & Technical Asset Valuation



IFRS 16 LEASE RECOGNITION
Right-of-Use (RoU) asset recognition for leased contractor fleets with automated interest & amortization schedules.

REVALUATION & IMPAIRMENT
Integrated impairment testing pursuant to IAS 36 when ore grade decreases or reserve estimates are adjusted.

DECOMMISSIONING PROVISIONS
Capitalized site restoration asset with unwinding interest provisions under SYSCOHADA Class 19 provisions.

05 Fleet & Plant Maintenance Management

In open-pit and processing operations, unscheduled equipment downtime halts downstream production, resulting in staggering financial losses. LedgerCore's Maintenance Engine enforces **Preventive, Corrective, and Telematics-Driven Service Schedules** directly connected to warehouse inventory and budget envelopes.

OPERATING HOURS TRACKING

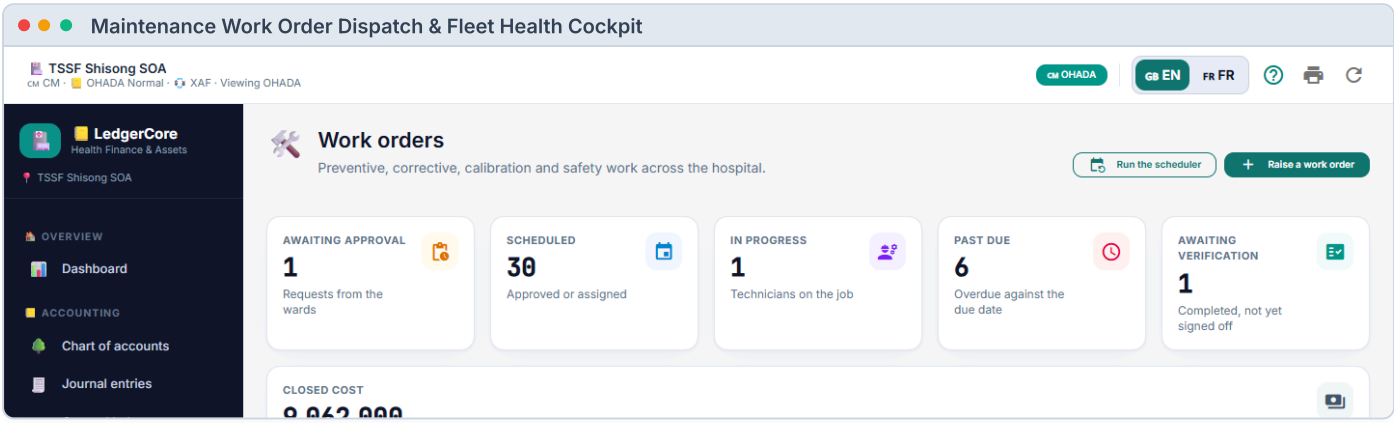
Continuous engine-hour logging from telematics and pre-shift inspections. Automated dispatch of Service PM-1 (250h oil/filters), PM-2 (500h hydraulics), PM-3 (1,000h transmission), and PM-4 (2,000h major engine overhaul).

WORK ORDER LIFECYCLE

End-to-end work order tracking: Fault reporting → Technician assignment → Parts reservation → Workshop execution → QA Supervisor inspection → Commissioning sign-off and historical log archiving.

CRITICAL SAFETY CHECKLISTS

Integrated calibration and safety governance for weighbridges, laboratory spectrometry scales, pressure vessels, and hoisting cables. Restricts machine dispatch if statutory safety testing lapses.



Mean Time Between Failures (MTBF) & Spare Parts Dual Validation

When a work order is generated for a major overhaul (e.g., replacement of excavator hydraulic pumps), the system automatically performs dual validation:

1. Warehouse Spare Parts Reservation:

Locks critical parts in warehouse stock to ensure parts are not double-allocated. Deducts stock upon work order sign-off and posts consumption directly to SYSCOHADA Class 31/60 accounts.

2. Budget Ceiling Check & Commitment:

Instantly checks the pit's maintenance budget allocation. If sufficient credits exist, a real-time statutory commitment is registered, preventing unauthorized expenditures.

06 Budgeting & Statutory Commitment Accounting

Public authorities and international joint-venture partners in the extractive sector demand absolute budgetary discipline. LedgerCore embeds a complete **Multi-Axial Budgeting and Real-Time Commitment Accounting (Comptabilité des Engagements)** engine designed after sovereign West African institutional standards.

📊 Multi-Axial Budgetary Architecture

Budgets are structured across four simultaneous analytical axes:

- ✓ **Concession / Mining Pit:** Concession North, Pit Alpha, Tailings Dam, Processing Plant.
- ✓ **SYSCOHADA Account:** Class 6 (Opex: Reagents, Diesel, Maintenance, Salaries) and Class 2 (Capex: Mine Development, Heavy Fleet).
- ✓ **Department / Cost Center:** Drilling & Blasting, Haulage, Metallurgy, Safety & Environment, Administration.
- ✓ **Funding Program / JV Partner:** Own Equity, Commercial Debt, Joint Venture Shareholder accounts.

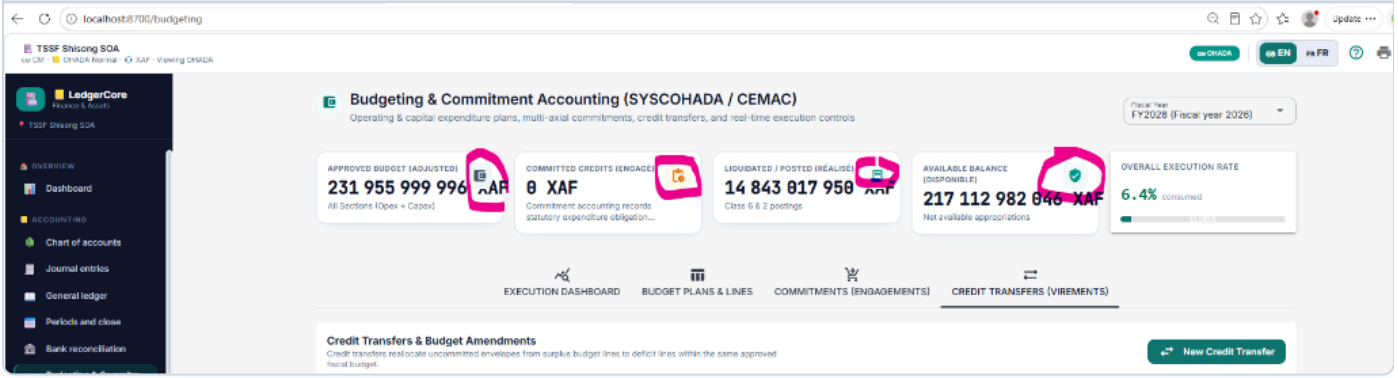
🛡️ Real-Time Overrun Blocking (Contrôle des Engagements)

In traditional ERPs, budgets are evaluated post-facto during invoice entry—when the liability has already been incurred. LedgerCore enforces **Pre-Commitment Hard Blocking**: When a purchase order, contractor agreement, or major work order is submitted, the engine checks:

```
Available = Revised Dotation - Actual Spent - Pending Commitments
```

⛔ **If Request > Available Balance: Transaction is strictly blocked.**

LedgerCore Budgeting & Commitment Interface — High-Visibility XAF Controls



VIREMENTS DE CRÉDITS (CREDIT TRANSFERS)

Formal decision-making workflow to transfer surplus appropriations from one operational line to another (e.g. reallocating exploration savings to heavy haulage maintenance) with dual-executive sign-off and complete justification audit history.

MULTI-CURRENCY CONTRACT RESERVATIONS

Reserve multi-year drilling contracts in foreign currency (USD / EUR) with automatic conversion to functional currency XAF and variance monitoring against foreign exchange fluctuations.

07 Seamless Cross-Module Synergy: The Closed-Loop Workflow

The crowning achievement of the LedgerCore platform is the **absolute, zero-latency integration** among Accounting, Asset Management, Fleet Maintenance, and Budgeting. The following real-world mining scenario illustrates this closed-loop harmony:

Scenario: 100-Ton Haul Truck CAT 777G Engine Overhaul
Location: Concession Pit Alpha | Asset Tag: EQ-CAT-777G-04 | Trigger: Telematics Hour Counter reaches 2,000 Operating Hours

1. **Telematics Trigger & Work Order Dispatch** MAINTENANCE
   The telemetry interface detects engine operating hours crossing 2,000 hours. The system automatically opens Preventive Work Order **#WO-2026-0842**, attaches the engine overhaul technical protocol, schedules workshop bay 3, and drafts the required parts list (gaskets, pistons, synthetic oils).

2. **Real-Time Budget Ceiling Verification & Commitment** BUDGETING
   The overhaul requisition (est. 18,500,000 XAF) automatically queries the *Concession Pit Alpha Heavy Fleet Maintenance* budget line. The engine confirms available headroom and creates an active statutory commitment **#BC-9402**, instantly decrementing available credits and locking the funds.

3. **Componentized Asset Lifecycle & Meter Update** ASSET MANAGEMENT
   Technicians scan the CAT 777G metal QR code. The system logs the replacement of Engine Component Serial #ENG-9982, updates the asset technical record, recalculates the remaining asset useful life under IAS 16 / SYSCOHADA, and resets the PM-4 counter to zero.

4. **Automated SYSCOHADA Journal Entries & Inventory Relieving** CORE ACCOUNTING
   Upon QA supervisor sign-off, the transactional posting engine automatically generates balanced double-entry vouchers: Debiting **Account 6241 (Equipment Maintenance)** or **Account 231 (Asset Capital Enhancement)** and Crediting **Account 311 (Warehouse Spares Inventory)** and **Account 401 (Vendor Payable)** with full VAT and withholding tax tracking. Zero manual bookkeeping required!

OPERATIONAL EVENT	MAINTENANCE MODULE	BUDGETING MODULE	ASSET MODULE	ACCOUNTING LEDGER
Hour Threshold Reached	AUTO-GENERATE WO	CHECK DOTATION	LOG OPERATING METER	IDLE
Parts Requisition	RESERVE STOCK	COMMIT FUNDS (#BC)	LINK PART TO ASSET TAG	PENDING VOUCHER
Work Order Execution	LOG LABOR & QA CHECK	HOLD RESERVED AMOUNT	UPDATE COMPONENT S/N	WIP RECORDED
Final Sign-Off	CLOSE & CERTIFY WO	LIQUIDATE COMMITMENT	UPDATE CARRYING VALUE	AUTO-POST SYSCOHADA

08 Security, Governance & Concession Rollout Roadmap

Deploying an enterprise system across operating concessions requires a structured, risk-mitigated implementation methodology. LedgerCore has been deployed in demanding industrial environments with zero operational disruption.

🔒 Mining Enterprise Governance Matrix

- ✓ **Zero-Trust Architecture:** TLS 1.3 encryption across all network tiers and automated session timeout on unattended terminals.
- ✓ **Cryptographic Fiscal Locking:** SYSCOHADA periods are sealed using SHA-256 digital signatures, ensuring complete immunity against retroactive tampering.
- ✓ **Complete Traceability:** Every asset movement, meter reading, work order sign-off, and credit transfer logs the user ID, timestamp, IP address, and previous values.
- ✓ **Disaster Recovery & Redundancy:** Automated hot standby replication with Point-In-Time Recovery (PITR) guaranteeing zero transactional loss.

🚀 Turnkey 12-Week Implementation Plan

Weeks 1–3: Scoping & Chart of Accounts Harmonization
Mapping existing mining cost centers, pits, and SYSCOHADA revised accounts.

Weeks 4–6: Master Asset & Fleet Ingestion
Tagging heavy machinery, importing historical depreciation & component records.

Weeks 7–9: Maintenance & Budget Pilot
Workshop terminal rollout, telematics configuration, and commitment ceiling rules.

Weeks 10–12: Concession Go-Live & Corporate Sign-Off
Parallel run validation, four-eye audit verification, and full production cutover.

Conclusion & Next Steps

LedgerCore provides mining operators with the certainty, control, and statutory compliance required to thrive in the CEMAC region. By eliminating the disconnect between pit maintenance, asset valuation, and financial ledgering, your enterprise gains an unbeatable operational advantage.

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